

Naveen S

Bengaluru, India — naveenselvaraj.selva@gmail.com — +91 89511 12467
linkedin.com/in/naveen-selvaraj- — github.com/nh-44



Profile

Software Engineer-in-Training with experience in **backend systems, document intelligence, applied machine learning, and GenAI tooling**. Proven ability to build data processing pipelines involving PDF parsing, semantic similarity, and LLM-based reasoning. Demonstrated leadership across innovation projects, hackathons, and IEEE-published research. Seeking roles in AI-driven document processing and backend systems.

Education

PES University, Bengaluru	2023 – 2027
B.Tech in Computer Science Engineering	CGPA: 7.80
Christ Academy Junior College	2021 – 2023
2nd PUC	96.66%
ACTS Secondary School	2009 – 2021
SSLC	94.66%

Technical Skills

Skills Summary: Python, NLP, Document Processing, Semantic Similarity, OCR, LLMs, REST APIs, PostgreSQL, AWS, Git

Languages: Python, C, C++, JavaScript

Backend & Web: Node.js, Express, React, Next.js, REST APIs, RBAC

Databases: PostgreSQL, MySQL, Supabase

ML & AI: Scikit-learn, Random Forests, Vision Transformers (BEiT), XAI

Cloud & DevOps: AWS (Cloud Practitioner), Amazon Bedrock, Amazon Q, GitHub Actions, CI/CD, Jira, Docker & Kubernetes

Tools: Git, Streamlit, MATLAB, Simulink, Fusion 360

Productivity: Microsoft Excel, Word, Canva

Projects & Research

PatentEase (Capstone Project)

- Developing an end-to-end system for patent officers and inventors to analyse patent documents.
- Building a document processing pipeline for extracting and comparing patent content using semantic similarity and embeddings.
- Implementing AI-based novelty detection, summarisation, and multilingual translation of technical documents.
- Exploring LLM-based approaches for intelligent comparison and interpretation of patent data.
- Working on building a logic engine to identify novelty and predict patentability score.

LifeTag – Heal-O-Code Hackathon (1st Place)

- Designed an NFC-based emergency medical ID system enabling real-time access to critical health data across 200+ users.
- Implemented role-based access for First Responders, Doctors, and Users with privacy-first design.
- Enabled dynamic URLs, emergency-only data views, scan history, and breach detection mechanisms.

Medicinal Leaf XAI Classifier – IEEE INDICON 2025

- Summer Research Intern (2 months) at CoDMAV – Centre of Data Modelling, Analytics and Visualization, PES University.
- Researched Vision Transformers and Explainable AI for medicinal leaf classification.
- Built a multimodal XAI system integrating BEiT, TF-IDF, Grad-CAM, LIME, SHAP, and LRP.
- Published and presented the work at **IEEE INDICON 2025**.

Fuel Injection System Digital Twin – HAL Aerothon (Top 10)

- Worked with the team for 30+ hours developing a Simscape-based digital twin and ML-driven engine health monitoring pipeline.

Simplified GDPR Consent Manager (SGCM)

- Backend and QA engineering for a GDPR-compliant consent platform with RBAC, encryption-at-rest, CI/CD pipelines, and TDD.
- Worked with structured data workflows and system pipelines aligned with document processing systems.

HotShot (Personal Project)

- Real-time audience polling platform built using Next.js, Supabase, and analytics dashboards.
- Built XAI pipelines using Grad-CAM, LIME, SHAP for model interpretability.
- Enables participants to add options dynamically, improving flexibility over traditional polling systems.

Experience

Research Intern – CoDMAV (PES University)

- Worked on multimodal AI systems combining Vision Transformers and NLP.
- Built XAI pipelines using Grad-CAM, LIME, SHAP for model interpretability.
- Contributed to research published at IEEE INDICON 2025.

Club Head – Team Vegavath (Student Innovation Club)

- Revived and scaled the club into a leading innovation community on campus.
- Shifted focus to SDV, robotics, and automation systems.
- Organized **Ignition 1.0** (Ather Energy) — 230+ participants, 35-member team.
- Conducted second hackathon (Feb 2026) with ECE Dept and Xylem India.
- Led go-kart restoration, sponsorships, and cross-functional operations.

Teaching Assistant – PES University

- Assisted in labs, mentoring, evaluation, and academic coordination (MPCA).

Data Analytics Head – Nexus AWS

- Worked with AWS services and conducted cloud workshops.
- Explored GenAI tools (Amazon Bedrock, Amazon Q).

Professional Development Director – Rotaract PESU ECC

- Organized skill-building workshops and campus collaborations.

Member – PAWS

- Volunteered in animal welfare drives and fundraising initiatives.

Class Representative (3 Years)

- Managed student–faculty coordination, communication, and logistics.

Achievements

- 1st Place – Heal-O-Code (18 Hour Hackathon - 2025)
- Top 10 – HAL Aerothon (24 Hour Hackathon - 2025)
- Top 5 – Space Race Ideathon (2024)
- 8th Position – Get Fit Bharat (All India Cycling Event)
- IEEE INDICON 2025 – Research Publication

Interests

Endurance cycling, swimming, strength training, automotive engineering, film and TV analysis